

XXIII Indian Antarctic Expedition Hydrographic Survey—India Bay

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INTRODUCTION

The Hydrographic Survey of India Bay, Antarctica HI/NHO/01/2003-04 was undertaken during the XXIII IA expedition from 05 Dec 03 to 30 Mar 04. A total of 310 nautical miles of sounding was carried out by the expedition team. Apart from the bathymetric survey, current, salinity and sound velocity observations were also carried out during the expedition at various places. CTD probe was employed for data collection for the first time during the survey. The 150 NM ice shelf around the India Bay was delineated using System GPS 200 on Kinematic in the Fly mode. Meteorological observations were also carried out during the survey.

GEODETTIC CONTROL

Horizontal control for the entire survey was established in WGS 84 Datum and plotted on Mercator Projection with scale true at 66° S. GPS coordinates as directly obtained from the ship fitted GPS “Trimble Navtrac XL” were used for sounding. No control was established for this purpose. However, the output of the ship fitted GPS set was compared with the Leica GPS 200 available with the survey team and positions were found to be acceptable. The following existing station was recovered and used for extension of Geodetic Control. The coordinates of the station in WGS 84 Datum are as follows:

Stn.Id	Station Name	Latitude Longitude	Ell Height (in M)	Source
A	Maitri	S 70° 45' 51".730 E 11° 44' 02".570	117.068	HI/NHO/01/2003-04

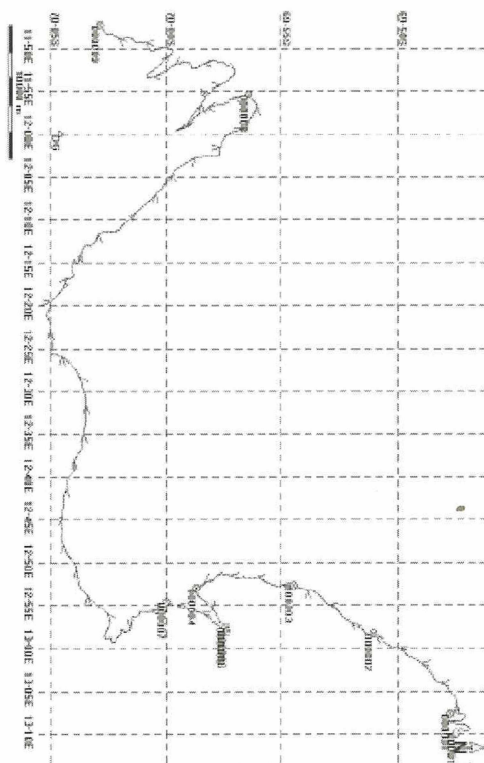
BATHYMETRY

The bathymetric survey was undertaken using the ship fitted echo sounder. Two Russian made echo sounders were available onboard. Model M4 designed for measuring depths up to 35 m and model NEL M3B for measuring depths up to 500 m. These instruments are analogue in nature and have no digital output to log the data in an Automated Survey System. In view of this, the data recorded was manually extracted from the analogue echo trace using manually graduated scale and corrected for draft and calibration. The sound velocity was fixed at 1500 m/s on the echo sounder and used for depth data collection. As all the soundings were more than 200mtrs, they were then corrected for index error of the echo sounder, ships draught and sound velocity using the Nautical Publication 139 echo sounder correction tables for Antarctic region. The sounding lines were run at approximately 1 cm apart on scale the scale of survey (1 : 200,000), perpendicular to the depth contours. A cross line was also run in the area. A total of 310 Nautical miles of sounding was completed in areas designated as A, B and C on chart INT 9051.

COASTAL TOPOGRAPHY AND TOPOGRAPHICAL FEATURES

Delineation of the complete ice shelf was carried out on 15 Mar 04. The ice shelf was surveyed using the integral helicopter at an average speed of 80 knots and height of 200 feet was maintained during the operation. The data was logged on Leica GPS 200 in Kinematic on the Fly mode. The GPS data of the IGS Station, DAVIS (Australia) was used as reference for processing the logged data. This station was used for baseline measurements to the ice shelf. The entire GPS data was processed in baseline mode using SKI version 2.3. The graphical representation of the complete ice shelf from 11° E to 14° E was plotted on a scale of 1:50 000 (enclosed figure) for studying the changes in the ice shelf. The coast or the ice shelf is totally barren and desolate. The ice cliff forms the seaward edge of the ice shelf. In most locations it is about 10-15 metres above waterline. Extreme caution needs to be exercised while moving on the shelf by foot or motorized vehicle. These cliffs are extremely dynamic features and are prone to breaking off and separating. Although they paint distinctively on radar, they become visually distinct only on melting of the fast ice in the summer. There is no plantation or vegetation even during the summer. The only feature that at times is visual during the summer months are some crevasses. There are no conspicuous objects along the coast falling within the survey area. However, there are numerous containers lying at the Indian and Russian

Discharge Points on the ice shelf at Indian and Russian Bay respectively. These can be used as general guideline, but not for any position fixing, since the containers are moved by the Expedition members for route marking or back loading as and when the requirement arises. Graphical representation of the newly acquired ice shelf data is shown below :



5. Comparison with Published Chart and Previous Surveys

Area 'A' and 'C': During the course of survey, it was found that the 500 meter contour in area 'A' has shifted south wards by approximately 5.0 miles in the centre of the area in position $69^{\circ} 35' S$, $11^{\circ} 12' E$. However, the contour in position $69^{\circ} 38' S$, $11^{\circ} 00' E$ and $69^{\circ} 31' S$, $11^{\circ} 25' E$ is matching with the existing contour on international chart INT 9051. In Area 'C' there is a slight shift in the 500 meter contour in North ward direction by about 1 NM in position $69^{\circ} 31' S$, $11^{\circ} 29' E$. In the rest of the area, the contour matches with the existing chart. Soundings in the area are in general agreement with those depicted on chart.

Area 'B' : The bathymetric data collected in area 'B' is in general agreement with the existing one. There are no contours in the area and as such no shift could be ascertained.

Ice shelf: The entire ice shelf from 11° E to 14° E for about 150 NM was demarcated using the expedition helicopter. The changes in the ice shelf are as follows:-

- a) The shelf at India Bay has lost a tongue in the eastern side, along with some portions of Leningrad Bay.
- (b) The shelf has extended about 2 miles seawards in area between 11° 00° E and 11 ° 20° E.

MISCELLANEOUS OBSERVATION

Sound Velocity was observed during days of sounding using EG&G Smart acoustic Current Meter. Tidal stream observations were also carried out using the above instrument. CTD was employed for the first time during this expedition and the data logged has been forwarded to NHO. The CTD was also employed during the passage to Eastern Antarctica and data collected. The NHO team also assisted the Special Task Force constituted by the Government of India for selecting the suitable area for a new station in eastern Antarctica, by carrying out CTD observations in the lakes of Rauer group of Islands and at various other places. Apart from this Meteorological data such as dry and wet temperatures, sea state and swell wind speed and direction etc was also collected by the Naval team.

OBSERVATIONS AND RECOMMENDATIONS

The special Task Force setup by Govt. of India to determine the site for new Indian station in east Antarctica was appraised by the NHO team for undertaking detailed bathymetric survey of the proposed area. The survey is to be undertaken by a suitable survey motor boat using portable echo sounder and positioning system on large scale. The existing chart for the area is on very small scale and no sounding detail is available for approaching the proposed site by the ship. In view of this , it is proposed that the survey be carried out on scale 1 : 20 000 in the area near the Larsemann hill region as this will enable the expedition vessel and the barges to approach close to the proposed site for discharge of cargo and equipment.

CONCLUSION

The XXIII Antarctic Expedition was unique in many ways. It was the longest summer expedition totaling 104 days out of port. During this expedition, the vessel transited to eastern Antarctica (78° E) where the Special Task Force carried out reconnaissance for a suitable site for the New Indian Station. The NHO team was successful in undertaking most of its assigned tasks. It completed 310 NM of ship sounding in three different areas and undertook 150 miles of coastlining from 14° E to 11° E longitude using the integral helicopter. In addition, physical oceanographic data, meteorological observations and synoptic system studies in the Antarctic continent, confirmation of Co-ordinates of Geodetic station at 'Maitri' and CTD data was collected. The team maintained a very healthy rapport with all the other expedition members and the ship's crew. Shano Varuna, Jai Hind.